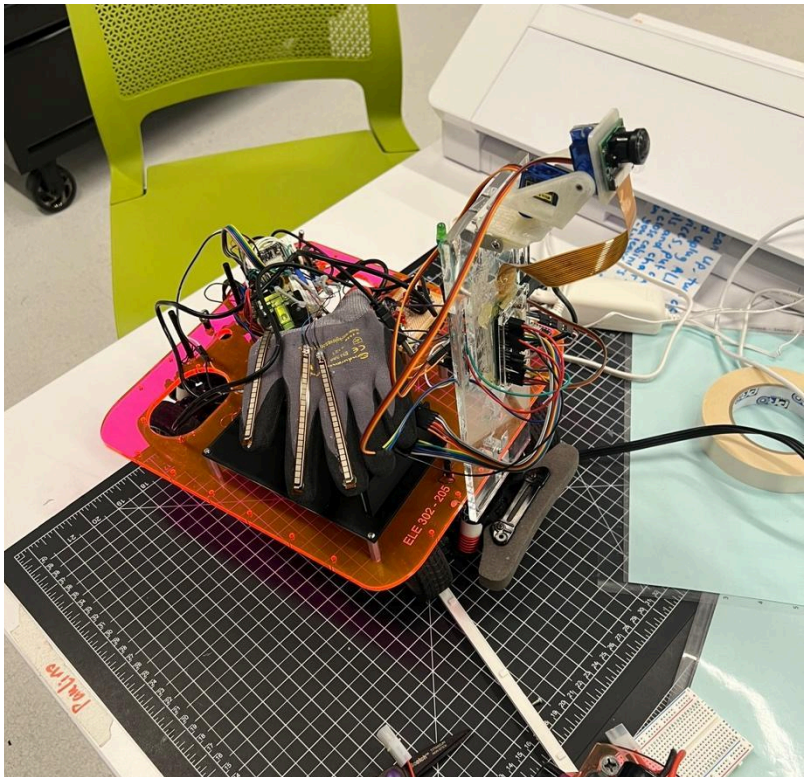
Brian Goodluck Mmari
ECE302
12/14/2023

FINAL PROJECT LAB REPORT

STATEMENT OF OBJECTIVE

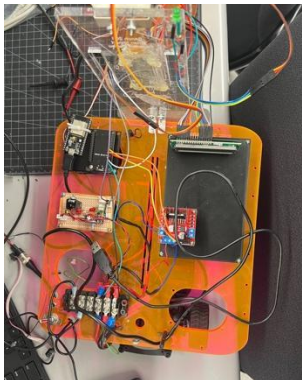
The objective of this project was to redesign the autonomous car as well as make a left-handed glove that can help the user to be able to control the car's motion. The glove, however, can only be used to control the direction of the car as well as the current state of the car(stationary/motion) but not change the speed of the car while in motion.

KEY SUB-SYSTEMS AND COMPONENTS



From the diagram above, the following are the main components and their subsystems:

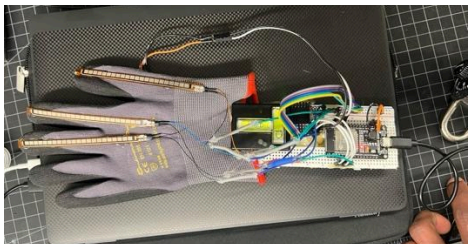
1. Car



- **H-Bridge:**
  - For our car, I needed to add the extra functionality where the car would be able to move both forward and backward. H-bridge motor drivers offered this extra functionality by wiring it directly to the already-existing motor of the car and reversing the direction of current through the motors.
- **Power Board:**
  - **Voltage regulator, 7805**: This is a voltage regulator chip that helps output 5V no matter the input voltage. The output of this voltage regulator is fed into the ESP32.
  - **Voltage regulator, 7806**: This is a voltage regulator chip that helps output 6V no matter the input voltage. This was left from previous projects just in-case we needed an extra power source to use.
  - **LED**: It is good practice to always have LEDs so as to ensure that the power board is being fed power and also enable easy troubleshooting when something isn't working.
- **ESP32 Board:**
  - **2 ESP32's:** The purpose of the ESP32 is to provide wireless communication between two objects.
    - The 1st ESP32(Car ESP32) was to be used as the main controller for the car. ESP32 was favoured because of its ability to act as a controller for the car as well as act like a 'receiver' for the wireless signals sent by the ESP32 from the glove. Everything was coded in Arduino IDE and then uploaded to the ESP32.
    - The 2nd ESP32(Wi-Fi ESP32) was to be used to connect to a self-built web server. This was an extra feature to enable connection to a personally built-Wifi (named 'Isabella') and enable the user to move the Raspberry Camera either from the comfort of his phone or laptop while moving the car with the gloves.
- **Raspberry Pi Zero + Raspberry Pi Camera + LCD Display:**
  - **Raspberry Pi Zero**: Offers the ability to power the Raspberry Pi Camera as well as connect to our personal Wi-Fi("Isabella").
  - **Raspberry Pi Camera**: Offers the ability to video stream via your device(laptop/phone).
  - **LCD Display**: This displayed the current IP address and Port number of the web server, which were prompted whenever the user tried to access the custom-built web server.



2. Glove

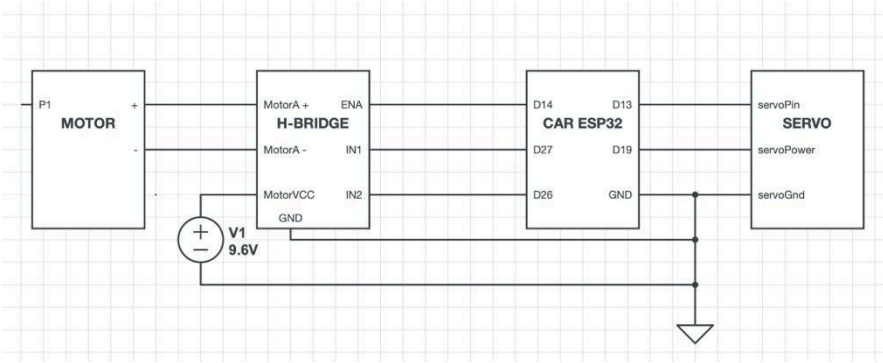


- **Breadboard:**
  - This was done to make the connections easier for both the Glove ESP32 as well as the wiring of the flex sensors.
- **The 2nd ESP32(Glove ESP32):**
  - This was used as the main controller of the gloves as well as acting like a 'transmitter' by sending wireless signals to the ESP32 of the car. Everything was coded in Arduino IDE and then uploaded to the ESP32. This can be powered either by a mini-USB cable to the laptop or to a rechargeable battery.
- **Flex Sensors + Pressure Sensors + Spirit level:**
  - The flex sensors work by capturing the change in resistance as someone flexes/bends their fingers, thereby sending analog resistance readings to the ESP32 of the car which was then coded with certain threshold values to determine what direction is left/right or backward.
    - Flex Sensor on the middle finger when bent enables the car to move to the right.
    - Flex Sensor on the index finger when bent enables the car to move backwards.
    - Flex Sensor on the thumb when bent enables the car to move to the left.
  - The Pressure Sensors put on the palm of the glove work by capturing the amount of pressure pressed on it, converting it to electrical signals and then sending analog electrical signals to the ESP32 of the car which was then coded to enable the car to move forward when pressed and stationary when not pressed.

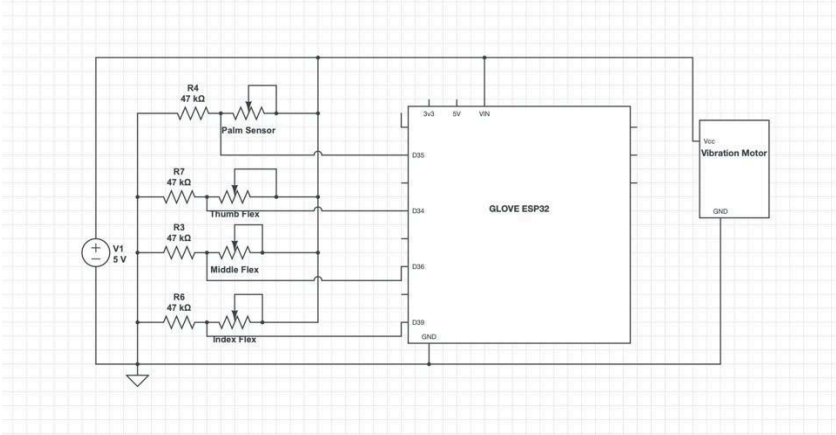
SCHEMATIC DIAGRAM OF CIRCUITS

OVERVIEW SCHEMATIC:

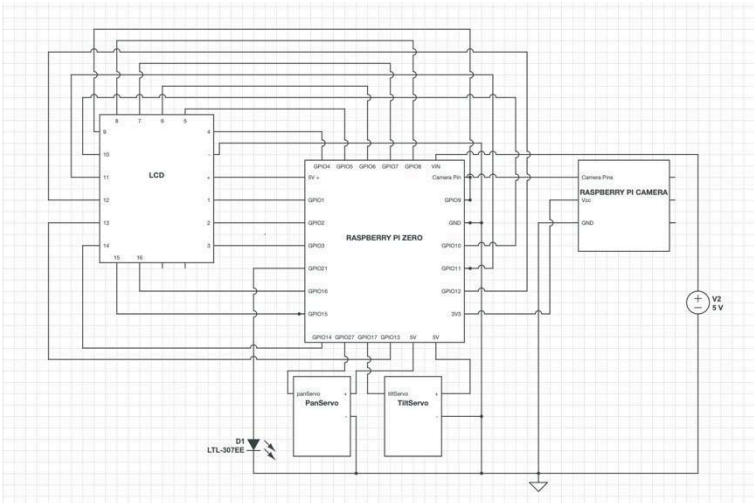
(1) Car Schematics



(2) Glove Schematics



(3) Raspberry Pi Zero + Raspberry Camera + LCD Schematics



CHALLENGES & DESIGN CHOICES MADE

(1) Car Weight Affecting Speed

The car was heavier than expected due to the use of rechargeable batteries, which were necessary for some components that required USB ports. This additional weight caused the car to move slowly, even at the maximum PWM signal of 255. However, when tested on a stand (without ground friction), the car moved effortlessly.

(2) Car Unable to Turn Left
We encountered an issue where the car was unable to turn left. The root cause was unclear but could have been due to either a loose connection or a damaged flex sensor. Further testing and troubleshooting would be required to confirm the exact cause.

(3) Difficulties with XBee Communication
Setting up XBees for wireless communication proved challenging. Initially, we attempted to configure two XBees using XTCU software, assigning one as a coordinator and the other as an end device (transmitter & receiver). However, despite multiple attempts, communication failed. As a result, we decided to switch to ESP32 for wireless communication, which was more reliable and easier to configure.

(4) Electrical Sparks and Loose Connections
We observed sparks between screws and some of the electronic boards mounted on the car. This was likely caused by loose wire connections, increasing the risk of short circuits. A more secure wiring approach, such as using heat shrink tubing, twisted-pair wiring, or direct soldering, could have significantly improved connection reliability. In cases where jumper wires were not the best solution, alternative wiring methods should have been used to enhance safety and durability.

WHAT COULD BE IMPROVED?

(1) Improved Mounting for the Raspberry Pi Zero
Instead of gluing the Raspberry Pi Zero to the mast, a more elegant and secure solution would have been to 3D-print a mounting board with drilled holes for screws. This would not only improve the car's aesthetics but also provide better stability and durability for the Raspberry Pi.

(2) Enhancing the Glove's Design
The glove could have been designed to be more visually appealing and structurally organized. Using a 3D-printed hand cover to mount and integrate all components together would have improved both the look and functionality of the glove by keeping everything neatly in place.

(3) Consistent Wire Color Coding
Maintaining standardized wire color coding would have improved debugging and organization:

- **RED/YELLOW** → Power
- **BLACK/GREEN** → Ground
  This practice ensures clearer wiring identification and reduces potential connection errors